Python Programming		
Course Code	**Course Name**	**Credits**
IFM4112	**Python Programming**	**04**

Contact Hours			**Credits Assigned**			
Theory	**Practical**	**Tutorial**	**Theory**	**Practical**	**Tutorial**	**Total**
03	-	01	03	-	01	04

Theory						**Term Work / Practical/Oral**			
Internal Assessment				**End Sem Exam**	**Duration Of End Sem Exam**	**Term Work**	**Pract.**	**Oral**	**Total**
Test	**Continuous Evaluation**	**Attendance**	**Total Internal**						
15	**10**	**05**	**30**	**70**	**3 Hours**	**-**	**-**	**-**	**100**

Course Outcomes

CO1: Install and configure the Python programming environment to develop and execute Python applications.

CO2: Develop Python programs using variables, data types, operators, input/output statements, and control structures.

CO3: Apply Python collection data types such as lists, tuples, dictionaries, sets, and ranges to solve programming problems.

CO4: Design and implement modular Python programs using functions and object-oriented programming concepts, including classes and objects.

CO5: Implement file handling and exception handling techniques to build robust and reliable Python applications.

CO6: Develop complete Python applications by integrating programming concepts, data structures, functions, and object-oriented principles to solve real-world problems.

Course Objectives

1. Understand Python Fundamentals: Understand the fundamental concepts of Python programming, including syntax, data types, operators, expressions, and control structures.
2. Develop Problem-Solving Skills: Develop problem-solving skills using functions, modules, and Python's standard libraries to create efficient programs.
3. Apply Object-Oriented Programming: Implement object-oriented programming concepts such as classes, objects, inheritance, polymorphism, and encapsulation in Python.
4. Perform Data Handling Operations: Perform file handling, exception handling, and utilize external libraries to build reliable and efficient Python applications.
5. Develop Real-World Applications: Design and develop real-world applications using Python in areas such as data analysis, web development, automation, and scripting.
6. Use Python for Data Structures and Algorithmic Problem Solving: Apply Python data structures (lists, tuples, dictionaries, sets) and algorithmic techniques to develop optimized solutions for computational problems.

Detailed Syllabus

Module / Unit	Course Module / Contents		Hours	Marks Weightage
	Module I : Python Basics		**08**	**20%**
1	1.1	Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language Declaration & Assignments: Preliminaries: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types		
	1.2	Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, The del Statement.		
	1.3	Dictionaries: Creating Dictionary, Accessing and Modifying key: value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, The del Statement		
	1.4	Tuples and Sets: Creating Tuples, Basic Tuple Operations, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Tuple Methods, Using zip () Function, Sets, Set Methods, Traversing of Sets, Frozenset()		
	Module II : Flow Control		**09**	**20%**
2	2.1	Python Decision Making: The if Decision Control Flow Statement, the if...else Decision Control Flow Statement, the if... elif...else Decision Control Statement, Nested if Statement Python Loops: The while Loop, The for Loop, The continue and break Statements		
	2.2	Function and Methods: Defining A Function, Calling A Function, Types of Functions, Function Arguments, Anonymous Functions, Global And Local Variables, Using Optional And Named		

		Arguments, Using Type, Str, Dir, and other Built-In Functions		
	2.3	Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, The Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules		
	2.4	Python Exception Handling: Exception handlers in Python, try – except clause with no exception, except clause with multiple exceptions, the try – finally clause		
3	**Module III : : Object-Oriented Programming**		**09**	**15%**
	3.1	Classes and Objects, Creating Classes in Python		
	3.2	Creating Objects in Python		
	3.3	The Constructor Method, Classes with Multiple Objects, Class Attributes versus Data Attributes		
	3.4	Encapsulation, Inheritance, The Polymorphism		
4	**Module IV : Network Programming & Data Processing**		**09**	**15%**
	4.1	Regular Expression Operations: Using Special Characters, Regular Expression Methods, Named groups in Python regular expressions, Regular Expression with glob Module.		
	4.2	Python Network Programming: Python Network Services, Defining Sockets, Socket Programs		
	4.3	Python Database Connections: Python Databases interfaces, DB-API, Benefits of Python Database Programming, Defining MySQL database, Database Operations.		
	4.4	Python Data Processing and Encoding: Representing CSV Files in tuples, Defining JSON, Dealing with JSON data.		
5	**Module V : Data Visualization & GUI**		**09**	**15%**
	5.1	Data Visualization using Matplotlib and Seaborn: Customizing Plots, Plotting 2D arrays, Statistical Plotting		
	5.2	GUI Development using tkinter: GUI Development, Standard attributed for GUI, methods for Geometry management.		
6	**Module VI: Django Introduction**		**08**	**15%**
	6.1	Django and its Advantages, Feature,		
	6.2	Django Web server, Django Architecture, MVC-MVT Pattern, Apps Life Cycle.		
	6.3	Creating App in Django, Django Admin App.		
Total			**52**	**100%**

Recommended Books:

No.	Book Title	Author(s)	Publisher	ISBN-13 / Edition
1	Beginning Programming with Python For Dummies	Fabrizio Romano	Wiley	978-1119091196 (2nd Ed.)
2	Python Projects	Laura Cassell &	Wiley	978-

		Alan Gauld		1118908662 (1st Ed.)
3	Head First Python	Paul Barry	O'Reilly / Shroff	978-1491919538 (2nd Ed.)
4	Beginning Programming with Python for Dummies	John Paul Muller	Wiley India	978-8126573829 (2nd Ed.)
5	Python Cookbook	David Beazley & Brian K. Jones	O'Reilly / Shroff	978-1449340377 (3rd Ed.)
6	Mastering Django	Nigel George	Packt Publishing	978-1783553723 (1st Ed.)
7	Django Web Development with Python	Samuel Dauzon, Aidas Bendoraitis, & Arun Ravindran	Packt Publishing	978-1787121386 (1st Ed.)